

02 | OVERVIEW



Exploring Climate Vulnerability in Coulibistrie, Dominica

Abstract

Coulibistrie is a village of about 423 people on the west coast of Dominica, tucked into a river valley between steep mountains and the Caribbean Sea. After Tropical Storm Erika in 2015 and Hurricane Maria in 2017, residents are still rebuilding. Researchers used community sessions and geospatial mapping to understand why some places struggle to recover and how neighbors, local businesses, and outside partners can build resilience together.

Introduction

Dominica, known as the Nature Island, is the largest and most northerly of the Windward Islands. Its mountains rise over 1,400 meters, and heavy rainfall feeds rivers, lakes, and waterfalls. In 2019, Dominica's prime minister announced the goal of becoming the world's first climate-resilient nation. Coulibistrie shows why that goal matters. The village sits at the mouth of the Coulibistrie River, which is both the heart of community life and a source of danger during intense storms. Farmers grow plantains, citrus, avocados, and ginger, while fishers use pots, spears, and fish aggregating devices. Through the Advancing Community Climate Resilience Project, researchers from Northeastern University and the University of the West Indies worked with residents to understand how repeated disasters shape a community.

How Do Scientists Study This?



FROM THE ARTICLE

Coulibistrie, set in a valley where the river meets the Caribbean Sea.

Image: ACCRP StoryMap, Global Resilience Institute.

The team applied the Integrated Resilience Enhancement System (I-RES), which combines community knowledge with indicator data. In June 2024, residents joined a co-development session to identify assets and challenges in four domains: physical and natural environment, economic functions, governance, and social capital. Analysts then built geospatial risk maps. The flood map layered digital elevation, land use, and distance to waterways. The landslide map was based on slope, because steep hillsides are more likely to fail. Researchers compared these maps with the locations of homes, schools, churches, and roads, and reviewed records from recent storms, including Erika, Maria, Philippe, and Beryl, to see how each event affected the village.

Why recovery gets harder with each storm

1 Storm strikes

Floods and landslides damage homes and roads

2 Savings drained

Families spend personal funds to rebuild

3 Loans blocked

Disaster-area status limits bank lending

4 Next storm

Hits before recovery is complete

Study the Process

Gather local knowledge

Residents described storm damage, recovery struggles, and the strengths that helped them cope.

Map flood and landslide risk

GIS maps used elevation, land use, waterways, and slope to show which areas are most exposed.

Overlay community assets

Homes, churches, the school, and roads were placed on the maps to see what is at risk.

What Did They Discover?

Coulibistrie ranks 10th most vulnerable of 66 communities in Dominica. Its valley location, river, and surrounding steep hills put it at high risk of both flooding and landslides. Erika's river flooding damaged homes, farmland, and water systems. Two years later, Category 5 Hurricane Maria destroyed homes, public buildings, and businesses. Power was not fully restored for close to a year, and residents relied on the river to bathe, drink, and wash. The main highway and two key bridges were damaged beyond use. After the storms, the village was labeled a special disaster area, which made bank loans for rebuilding very hard to get. Yet neighbors shared water, fuel, and food, and new local businesses selling honey and herbal tea emerged.

Coulibistrie at a glance

~423

residents in 163 households (2023)

10th

most vulnerable of 66 communities in Dominica

~1 yr

without full power after Hurricane Maria

One resident summed up the experience: "How exhausting to have to rebuild and rebuild." Repeated storms drain savings and slow full recovery.

02 | TAKEAWAYS

Takeaways

Place shapes danger

A river valley surrounded by steep slopes faces both floods and landslides during heavy rain.

Disasters compound

When storms strike before recovery is complete, damage and financial stress pile up.

Neighbors are resilience

Sharing resources in the first days after a storm helped the village survive until aid arrived.

Key Terms

Vulnerability

How likely a community is to be harmed by a hazard, based on its location, resources, and ability to recover.

Landslide

The movement of rock, soil, or debris down a slope, often triggered by heavy rain on steep ground.

Geospatial mapping

Using location-based data, such as elevation and slope, to create maps that show patterns of risk.

Compounding disasters

Hazards that strike one after another, so each new event hits a community that has not yet recovered.

Key Terms continued

Social capital

The relationships, trust, and networks that help people work together, especially in a crisis.

Activity

Recovery Timeline

Students build a timeline of storms that hit Coulibistrie and trace how each one affected homes, money, and daily life.

Materials: Long strip of paper or a digital timeline; Markers in three colors; Lesson text

1. Place Tropical Storm Erika (2015), Hurricane Maria (2017), Tropical Storm Philippe (2023), and Hurricane Beryl (2024) on a timeline.
2. Under each storm, add its effects on infrastructure in one color, the economy in a second color, and daily life in a third.
3. Draw arrows to show how damage from one storm made the next one harder to recover from.
4. Circle two community strengths that helped residents cope.
5. Propose one change, such as better loan access or landslide warnings, that could shorten recovery after the next storm.

References

Global Resilience Institute, Northeastern University, and University of the West Indies. Advancing Community Climate Resilience Project: Coulibistrie, Dominica (ArcGIS StoryMap).
<https://storymaps.arcgis.com/stories/1b49c1a74cab42b8b7897044cf8501d2>